

BrainScope — Intermediate progress report and open questions

Alejandro Bordon — Medical Image Processing (11763)

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Repository. <https://github.com/AlejandroB10/BrainScope> (public; you can read the README and browse the layout from there).

1. What the project is about

The plan is to implement the three objectives of the proposal on a single brain study made of a dynamic PET (`e_1_BRAIN_DINAMIC_COLINA`) and a MR T1 (`AX_3D_T1`): load and visualize the DICOMs, rigidly coregister the temporally averaged PET onto the MR, and segment the tumor visible on the last PET frame with one of the AI models suggested in the proposal. The architecture stays small on purpose: one Python module per objective, the course conda environment `medical-image-processing-11763`, and a bridge to 3D Slicer through the `mcp-slicer` MCP server for live qualitative inspection.

2. Where I am right now

I went a bit further than I had planned for the intermediate. Objective 1 is essentially complete and Objective 2 has a working coregistration pipeline. Objective 3 hasn't started yet.

Objective 1. `src/loading.py` reads both DICOMs, reshapes the flat (1692, 256, 256) `int16` PET array into (36 frames, 47 slices, 256, 256), computes the temporal mean and the last frame, and saves three artifacts under `docs/figures/`: a static composite of the median planes for the last frame and the temporal mean, the same composite for the MR T1, and a 36-frame GIF that sweeps the three median planes across time. The reshape order was confirmed against `FramePositionsVector` (every z position appears once per temporal frame) and validated visually by reloading the result into 3D Slicer through the MCP bridge: the temporal mean reads as a recognisable brain (Figure 1).

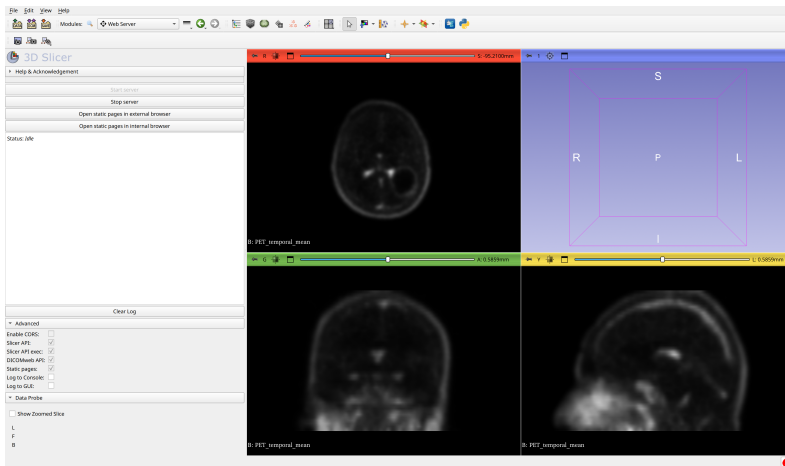


Figure 1: PET temporal mean reloaded into 3D Slicer through the MCP bridge. The volume is anatomically coherent and the metabolic activity is consistent with the lesion seen on the MR T1.

`src/coregistration.py` covers the **Objective 2** pipeline end-to-end. It implements a rigid model with translation plus axial rotation around the volume centre (Rodrigues), a Mutual Information loss computed from a 32-bin joint

histogram, and an optimisation driver that calls `scipy.optimize.minimize` with the Powell method. Running it on a downsampled (48, 64, 64) grid (the speed-vs-physics compromise I describe below) takes ~10 seconds and moves Mutual Information from 0.408 (identity init) to 0.526 (Figure 2).

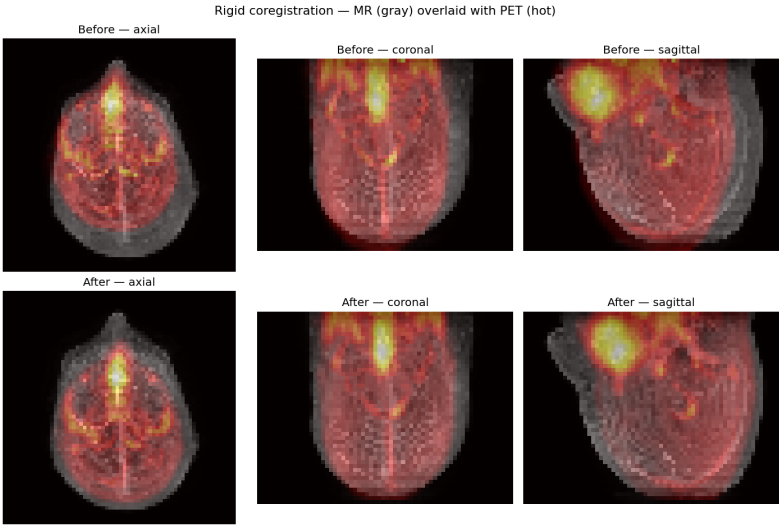


Figure 2: Rigid coregistration before (top) vs after (bottom). MR T1 in gray, PET temporal mean in hot. The PET footprint tightens onto the MR brain after the optimisation.

Objective 3 hasn’t begun yet. It’s parked behind the model choice, which I cover in the open questions.

A few things on the to-do list. The rotating-MIP animation of Objective 2.b is not yet done. The registration runs on voxel space after isotropic downsampling; the full pipeline must pull the PET / MR voxel-to-physical affines from the DICOM headers and resample on a common physical grid (top of the list after this submission). The Objective 3 mask propagation, the inverse transform, and the numerical coregistration assessment are all parked behind the Objective 3 work.

3. How I’m planning to attack the rest

The Objective 2 cleanup comes first: build voxel-to-physical affines from `ImagePositionPatient` / `ImageOrientationPatient` / `PixelSpacing` / `SpacingBetweenSlices`, resample PET onto the MR grid in physical coordinates, rerun the MI optimisation there. Once that lands, the rotating MIP animation drops out almost for free. Objective 3 then runs the chosen AI model on the MR side, prompted by a bounding box placed manually in 3D Slicer, and the resulting mask is propagated back to the PET space via the inverse rigid.

4. Risks I’m watching

Risk	Mitigation
Voxel-space registration hides physical mismatches	Switch to physical-coordinate registration before the final submission
Mutual Information local maxima	Multi-start initialisation; PyElastix as a sanity check
AI segmentation model too heavy for the local hardware	Pre-screen models; classical region growing as fallback
Time pressure on Objective 3	Keep the rotating-MIP polish minimal so the model work has runway

The list of open questions on the next pages is the input I need from you to lock those decisions before going further.

Open questions

These are the points where I'd like input before committing to an approach. Grouped by stage, ordered roughly by impact on the final grade.

Objective 1 — DICOM loading and visualization

1. **Slice ordering of the dynamic PET.** I'm using (0055, 1002) Frame Positions Vector, but the file also exposes `ImagePositionPatient` per slice. I confirmed against `FramePositionsVector` that every z position appears once per frame. In your experience, are there scanners where Frame Positions Vector and `ImagePositionPatient` disagree, and would you trust either one as the ground truth?
2. **Temporal aggregate for the coregistration input.** Right now I'm using a plain unweighted mean across the 36 frames. Frame durations range from 5 s to 300 s, so a duration-weighted mean would emphasise the late frames. Is the unweighted mean fine, or do you prefer the weighted one?
3. **Intensity scale for the GIF.** I lock `vmax` at the 99.5 percentile of the whole 4D volume and reuse it across frames so the early-frame wash-in vs. late-frame plateau remains visible. Does that match what you usually expect?

Objective 2 — 3D rigid coregistration

4. **Custom vs. PyElastix.** My implementation is from scratch (rigid + MI + Powell). I plan to keep it that way and only use PyElastix as a sanity check. Does that match the spirit of the assignment?
5. **Loss function.** I'm on Mutual Information with a 32-bin joint histogram. Would you push for normalised MI or normalised cross correlation on PET vs. MR specifically, or is plain MI good enough?
6. **Initial rotation prior.** I currently start from the identity (zero translation, zero rotation). For a same-session study this seems safe, but a PCA-based principal-axes alignment might bring the optimiser closer to the global maximum. Worth doing, or overkill?
7. **What to report numerically.** Right now I report MI before / after. What's the figure of merit you'd like to see in the final report? Final MI, residual landmark distances, target registration error from Slicer-placed fiducials, or a combination?
8. **Voxel-space shortcut.** The current pipeline registers in voxel space after isotropic downsampling. Lifting it to physical-coordinate registration is on my plate. Is doing both versions (voxel and physical) something you'd value as a discussion in the final document, or do you prefer just the physical one with the voxel one dropped from the report?

Objective 3 — 3D image segmentation

9. **Model recommendation.** Among `nnInteractive`, `SAMed-2`, `SAT`, and `MedSAM2`, which one would you pick as the default for brain tumors on PET / MR T1? Hardware-wise I have one consumer-grade GPU, which I think rules out the heaviest configurations.
10. **Modality the model sees.** Do you expect the model to run on the MR volume, on the coregistered PET, or on a fused channel? The decision really changes how I structure Objective 3.
11. **Reference mask.** Is there a ground-truth tumor mask available for this study? If yes, I'd compute Dice / Jaccard / Hausdorff. If not, I'll fall back to volume / sphericity / qualitative comparison and document the limitation.
12. **Bounding box workflow.** For the manual bounding box step, is a hard-coded tuple (taken once from a 3D Slicer inspection) acceptable, or do you want an interactive widget inside the report notebook?

General / submission

13. **Notebook vs. scripts.** Activities use `__main__` scripts. For the final 5-page summary, do you prefer a single Jupyter notebook that orchestrates everything, or modular scripts referenced from the doc?

14. **Reproducibility data.** The DICOMs are out of git because of size. Is a `data/README.md` with download instructions enough for the “publicly accessible” criterion, or should I host a sample subset?
15. **3D Slicer evidence.** I’m wiring up an MCP bridge to drive Slicer programmatically (already used to validate the PET reshape, see Figure 1). Does that count as the “third-party DICOM visualizer” Objective 1.b asks for, or do you expect manual GUI screenshots in the final report too?